

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0505

Red Hill 345 kV
December 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0505. The Interconnection Customer proposes a 839.7 MW natural gas combined cycle facility interconnecting at the Red Hill 345 kV substation in the Virginia area. The request was received on August 03, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 251 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,503,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Red Hill 345-138 kV TX is overloaded to 257.3% of its 506 MVA emergency rating for the loss of the Red Hill / Fluvanna P.S 345.0 kV. The project contributes 100.0% DFAX (839.7 MW) to this overload.

For the loss of Fluvanna P.S / Red Hill 345.0 kV, the Charlottesville / Mohican Substation 138.0 kV loads to 251.5% of its 168 MVA rating. Project DFAX: 30.9%.

For the loss of Red Hill / Fluvanna P.S 345.0 kV, the Barracks Rd / Mohican 138.0 kV loads to 218.0% of its 200 MVA rating. Project DFAX: 30.9%.

For the loss of Red Hill / Fluvanna P.S Substation 345.0 kV, the Timken Mercy / Barracks Road Substation 138.0 kV loads to 206.2% of its 223 MVA rating. Project DFAX: 30.9%.

Red Hill Substation / Timken Mercy 138.0 kV: 198.0% loading (390 MVA) following the outage of the Red Hill / Fluvanna P.S 345.0 kV; DFAX = 48.6%.

For the loss of Red Hill / Fluvanna P.S 345.0 kV, the Bluff City / Reach 138.0 kV line loads to 167.8% of its 229 MVA rating. Project DFAX: 17.7%.

For the loss of Fluvanna P.S / Red Hill 345.0 kV, the Rock Island Dam / Scottsville 138.0 kV loads to 147.5% of its 190 MVA rating. Project DFAX: 25.6%.

The Reach / Scottsville 138.0 kV is overloaded to 139.4% of its 188 MVA emergency rating for the loss of the Fluvanna P.S / Red Hill 345.0 kV L. The project contributes 25.6% DFAX (214.6 MW) to this overload.

For the loss of Red Hill / Fluvanna P.S 345.0 kV, the Rock Island Dam / Yankee Doodle 138.0 kV loads to 138.9% of its 217 MVA rating. Project DFAX: 25.6%.

Unisource Golden Valley / Unionville Hydro Project 2499 Ny 138.0 kV: 123.4% loading (266 MVA) following the outage of the Fluvanna P.S / Louisa Generation Facility 345.0 kV Line; DFAX = 5.2%.

The Yankee Doodle / Red Hill 138.0 kV is overloaded to 122.7% of its 270 MVA emergency rating for the loss of the Red Hill / Fluvanna P.S 345.0 kV. The project contributes 25.6% DFAX (214.6 MW) to this overload.

Crozet / Revital 138.0 kV line: 119.7% loading (167 MVA) following the outage of the Red Hill / Fluvanna P.S 345.0 kV; DFAX = 17.6%.

Grassmere Substation / Red Hill 138.0 kV Line: 117.6% loading (168 MVA) following the outage of the Red Hill / Fluvanna P.S 345.0 kV; DFAX = 25.9%.

For the loss of Fluvanna P.S / Louisa Generation Facility 345.0 kV, the Lake Lure / Bluff City 138.0 kV loads to 117.3% of its 480 MVA rating. Project DFAX: 8.8%.

For the loss of Fluvanna P.S / Red Hill 345.0 kV, the Boxelder / Timken Mercy 138.0 kV loads to 114.2% of its 265 MVA rating. Project DFAX: 17.6%.

The Boxelder / Video Pd 138.0 kV is overloaded to 113.3% of its 241 MVA emergency rating for the loss of the Fluvanna P.S / Red Hill 345.0 kV. The project contributes 17.6% DFAX (147.9 MW) to this overload.

For the loss of Red Hill / Fluvanna P.S 345.0 kV, the Mcauley / Video Pd 138.0 kV Line loads to 108.8% of its 228 MVA rating. Project DFAX: 17.6%.

For the loss of Red Hill 345/138 kV Transformer Bank, the Red Hill / Fluvanna P.S 345.0 kV loads to 107.2% of its 1,215 MVA rating. Project DFAX: 100.0%.

The Mcauley / Crozet 138.0 kV is overloaded to 105.7% of its 211 MVA emergency rating for the loss of the Red Hill / Fluvanna P.S 345.0 kV. The project contributes 17.6% DFAX (147.9 MW) to this overload.

For the loss of Fredericksburg / Polk City 345.0 kV, the Fredericksburg / Polk City 345.0 kV Circuit 1 loads to 104.9% of its 1,465 MVA rating. Project DFAX: 8.4%.

The Fredericksburg / Polk City 345.0 kV Ckt 2 is overloaded to 104.9% of its 1,465 MVA emergency rating for the loss of the Fredericksburg / Polk City 345.0 kV L. The project contributes 8.4% DFAX (70.5 MW) to this overload.

Gordonsville Energy Lp / Littlefork Substation 138.0 kV: 104.2% loading (457 MVA) following the outage of the Louisa Generation Facility / Fluvanna P.S 345.0 kV; DFAX = 6.4%.

Littlefork / Lake Lure 138.0 kV: 102.6% loading (593 MVA) following the outage of the Fluvanna P.S / Louisa Generation Facility 345.0 kV; DFAX = 11.7%.

The Birchwood / Dunn Siding 345.0 kV is overloaded to 102.3% of its 1,312 MVA emergency rating for the loss of the Leavitt / Birchwood Substation 345.0 kV line. The project contributes 7.1% DFAX (59.4 MW) to this overload.

For the loss of Fluvanna P.S / Louisa Generation Facility 345.0 kV L, the Fluvanna P.S 345 kV/138 kV Transformer loads to 102.0% of its 504 MVA rating. Project DFAX: 11.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Red Hill 345-138 kV TX	Replace transformer with higher-rated unit	\$20,333,000	63% (\$12,827,000)
Charlottesville / Mohican Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,185,000	59% (\$1,889,000)
Barracks Rd / Mohican 138.0 kV	Rebuild line with higher-capacity conductor	\$8,535,000	59% (\$5,061,000)
Timken Mercy / Barracks Road Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$5,286,000	59% (\$3,134,000)
Timken Mercy / Red Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$2,683,000	58% (\$1,545,000)
Bluff City / Reach 138.0 kV line	Rebuild line with higher-capacity conductor	\$17,420,000	46% (\$7,972,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rock Island Dam / Scottsville 138.0 kV	Rebuild line with higher-capacity conductor	\$19,251,000	62% (\$11,873,000)
Reach / Scottsville 138.0 kV	Rebuild line with higher-capacity conductor	\$7,957,000	62% (\$4,908,000)
Rock Island Dam Substation / Yankee Doodle 138.0 kV	Rebuild line with higher-capacity conductor	\$5,769,000	62% (\$3,558,000)
Unisource Glden Valley / Unionville Hydro Project 2499 Ny 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,531,000	58% (\$889,000)
Yankee Doodle / Red Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$5,928,000	62% (\$3,656,000)
Crozet / Revital 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,498,000	53% (\$2,364,000)
Grassmere / Red Hill 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,000,000	62% (\$1,866,000)
Bluff City Substation / Lake Lure 138.0 kV line	Rebuild line with higher-capacity conductor	\$24,951,000	17% (\$4,190,000)
Boxelder / Timken Mercy 138.0 kV	Rebuild line with higher-capacity conductor	\$2,942,000	53% (\$1,547,000)
Boxelder / Video Pd Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,306,000	53% (\$3,315,000)
Mcauley / Video Pd 138.0 kV line	Reconductor line	\$878,000	53% (\$462,000)
Red Hill / Fluvanna P.S 345.0 kV	Reconductor line	\$15,970,000	63% (\$10,074,000)
Crozet / Mcauley 138.0 kV Line	Reconductor line	\$2,103,000	53% (\$1,105,000)
Fredericksburg / Polk City 345.0 kV Cir. 1	Reconductor line	\$25,217,000	10% (\$2,582,000)
Fredericksburg / Polk City 345.0 kV Ckt 2	Reconductor line	\$31,673,000	10% (\$3,243,000)
Gordonsville Energy Lp / Littlefork Substation 138.0 kV	Reconductor line	\$753,000	13% (\$96,000)
Littlefork / Lake Lure 138.0 kV	Reconductor line	\$2,825,000	16% (\$454,000)
Birchwood / Dunn Siding 345.0 kV	Reconductor line	\$19,897,000	8% (\$1,700,000)
Fluvanna P.S 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$20,392,000	55% (\$11,193,000)
	Total allocated to project		\$101,503,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.